NIDAR AirMouse

Complete Problem Decomposition, Technical Subproblems & Problem-Specific Rules
Based on NIDAR-26-27 Rulebook v2.1 and Mission Brief – NIDAR AirMouse

Purpose: This document converts the official AirMouse requirements into clear engineering subproblems and collects the rules that directly affect the solution.

1. Overall Problem

The NIDAR AirMouse challenge requires an autonomous indoor drone to operate in a GPS-denied, maze-like environment. The drone must enter the arena, autonomously explore the unknown environment, detect up to six survivors, identify their grid locations, generate and continuously update a real-time 2D map, tag survivor locations on that map, and exit safely through the designated exit. The mission must be completed as quickly as possible while maintaining safety, autonomy, mapping accuracy and localisation accuracy.

> **Core mission flow:** Start → Enter maze → Explore → Localise → Map → Detect survivors → Determine grid locations → Tag survivors → Navigate to exit → Exit safely.

2. Engineering Subproblem Decomposition

#	Subproblem	What must be solved?
1	GPS-denied indoor localisation	Determine the drone's position without GPS/GNSS.
2	Autonomous navigation	Decide where to move next in an unknown maze.
3	Obstacle & corridor navigation	Safely fly through corridors, rooms, turns, junctions and obstacles.
4	2D SLAM / mapping	Build and continuously update a usable 2D representation of the explored environment.
5	Drone pose on map	Maintain the drone's estimated position within the generated map.
6	Survivor detection	Autonomously detect survivors using onboard sensing and processing.
7	Survivor grid localisation & tagging	Determine the correct 1 m × 1 m grid cell and automatically tag it.
8	Real-time GCS / Mission Planner	Display video, map, drone position, survivor tags and mission progress live.
9	Autonomous completion & safety	Reach/exit the designated point and finish safely without manual intervention.

Note: These nine items are an engineering decomposition created from the official requirements. They are not official NIDAR numbered subproblems.

3. GPS-Denied Indoor Localisation

GPS/GNSS cannot be used during mission execution. The drone therefore needs an onboard method to estimate its position and orientation. The system needs a useful estimate of position, altitude and heading so that navigation, mapping and survivor localisation can work together. The official brief does not prescribe a specific localisation algorithm.

- Key challenge: localisation drift in a visually repetitive indoor maze.
- The localisation system must support the generated map and survivor grid tagging.
- The solution should continue operating without an external GPS/GNSS source.

4. Autonomous Navigation

After mission initiation, the drone must autonomously enter the maze, navigate the indoor environment, identify corridors, turns, junctions, rooms and obstacles, detect survivors, map the environment, proceed toward the designated exit and exit safely.

• The exact internal maze layout may not be disclosed before the mission.

• This means the system should be capable of exploration rather than relying only on a fixed pre-known route.

• Navigation decisions should be generated by the autonomy stack rather than operator steering.

5. Obstacle & Confined-Space Navigation

The arena is a covered, maze-like indoor environment. The total arena size shall not exceed 15 m × 15 m. Corridors have at least 1 m clear width and at least 8 ft vertical clearance. The standard room size is 2 m × 2 m, and the height clearance is 8 ft.

• The drone must be configured to operate safely in corridors of the specified width.

• The drone must avoid touching walls, ceiling, arena panels and obstacles.

• Obstacle sensing and local avoidance are therefore core safety and autonomy functions.

6. 2D SLAM / Indoor Mapping

The drone must generate a real-time 2D map of the explored environment and continuously update it during flight. Mapping is not allowed to be postponed until after landing.

• The scoring evaluates correctly mapped 1 m × 1 m grid cells.

• For each grid cell, the reference evaluation considers walls, openings, corridors or accessible directions adjoining that cell.

• The map must be transferred to and displayed at the GCS during flight.

• There is no additional post-flight time for map generation, correction or transfer.

7. Drone Position on the Map

The GCS must display the drone position or estimated drone position within the mapped area. Therefore, localisation and mapping must be sufficiently integrated to maintain a meaningful relationship between the drone pose and the generated map.

• A map without a reliable drone pose is insufficient for the required mission display.

• Pose accuracy also directly affects survivor grid localisation.

8. Survivor Detection

Up to six survivors may be placed at different locations in maze or room-like sections. They may be real humans or dummies. The drone must detect survivors using onboard sensing and processing.

• Detection must be autonomous.

• Detection must occur during flight.

• The operator cannot manually identify/tag survivors through the GCS.

9. Survivor Grid Localisation & Tagging

Detecting a survivor is only part of the requirement. The system must identify the grid coordinate or grid box containing the survivor and place the survivor tag on the generated map.

• The scoring definition uses the 1 m × 1 m reference grid.

• A survivor earns the detection/localisation marks only when detection and correct map tagging are both achieved.

• Manual tagging or operator-assisted correction is considered manual intervention.

10. Real-Time GCS / Mission Planner

The GCS is a mandatory mission interface, not merely a video screen. It must display live mission status, live camera feed, the continuously updated 2D map, identified corridors/rooms where technically feasible, survivor grid information and tags, drone position/estimated position, and mission progress.

• All required mission data must be generated, displayed and transferred during flight.

• The operator's authorised supervision and live video viewing must occur through the designated GCS.

• FPV goggles and separate piloting/navigation/monitoring/video-viewing devices are prohibited.

11. Autonomous Exit & Mission Completion

The drone must proceed toward the designated exit point and exit the maze safely. The final mission gives 50 marks for successful autonomous exit without manual intervention.

• Maximum permitted flying time: 30 minutes.

• Fast-completion bonus threshold: within half of the maximum time, i.e. 15 minutes.

• Speed should not be achieved at the expense of mapping, localisation or safety.

12. Final Mission Scoring — 600 Marks

Criterion	Scoring method	Max
Survivor Detection & Localisation on 2D Map	40 marks for each correctly detected survivor whose location is correctly tagged on the generated 2D map. Correct	240
2D Grid Mapping Accuracy	Marks based on correctly mapped 1 m × 1 m grid cells. Evaluation considers walls, openings, corridors or accessib	220
GCS / Mission Planner Map Display	Yes/No. Full marks if GCS displays generated 2D map, estimated drone position, survivor tags and mission progres	50
Autonomous Exit	Yes/No. Full marks if the drone exits through the designated exit without manual intervention.	50
Fast Completion Bonus	Yes/No. Full marks if the mission is completed within half of the maximum permitted mission time.	25
Safe Mission Completion	Yes/No. Full marks if mission completes without crash, unsafe contact, uncontrolled landing or organiser interventio	15
TOTAL		**600**

Priority insight: Survivor detection/localisation + mapping account for 460 of the 600 Final Mission marks (76.7%). Therefore, these should be the highest-priority technical areas.

13. Manual Intervention Rules

• Manual control input is not allowed after mission start.

• Manual path correction is not allowed.

• Waypoint adjustment is not allowed.

• Operator-assisted navigation is not allowed.

• Manual survivor tagging is not allowed.

• Manual drawing/correction of the map is not allowed.

• Operator-assisted correction of survivor locations is not allowed.

• The operator may start the mission and trigger the safety abort/emergency stop when required.

FPV restriction: The operator may not use FPV goggles or another separate piloting, navigation, monitoring or video-viewing device. Authorised mission supervision and live video viewing must take place through the designated Mission Planner / GCS.

14. No Post-Flight Processing

Mapping, survivor localisation, data processing and transfer of mission data to the GCS must occur during flight. No additional time is provided after the flight for generating/completing the map, processing data, transferring data or correcting the GCS.

15. Communication & Network Rules

• The mission must be designed for a no-external-network environment.

• The system cannot rely on GSM, LTE, 5G, public Wi-Fi, Internet or cloud connectivity.

• Drone-to-GCS communication must use the team's local communication link or onboard autonomy system.

• External network-based communication during the mission is treated as a violation or external/manual intervention, as applicable.

16. Team & Operator Rules

• Maximum two team members may set up the drone, GCS, communication systems and associated equipment during the 5-minute setup period.

• No more than one operator may supervise the mission through the GCS.

• No other team member may assist during setup, launch, mission execution, mapping, survivor detection, exit, landing, troubleshooting or recovery.

• Other team members may only observe and may not provide verbal, physical, digital or technical assistance.

17. Drone & Arena Constraints

Requirement	Rule
Complete ready-to-fly airframe	Commercially available market-ready / ready-to-fly complete drone airframes are not permitted.
Maximum all-up weight	10 kg including battery, onboard systems, sensors, communication equipment, payload mechanisms and other com
Propeller protection	Propeller covers/guards must fully protect the propeller operating area.
Arena	Covered, maze-like indoor arena; maximum 15 m × 15 m.
Corridor width	Minimum 1 m clear width.
Vertical clearance	Minimum 8 ft; 8 ft in corridors and rooms.
Standard room	2 m × 2 m.
Grid	Modular grid structure; mapping is evaluated using a 1 m × 1 m reference grid.
Entry	Drone must begin from the designated entry point.
Launch area	Fixed 2 ft × 2 ft launch area; no part of the drone may cross/remain outside it before mission start.
Flight time	Maximum 30 minutes from take-off until maze exit or mission termination.

18. Safety & Failsafe Requirements

• Emergency stop / mission abort capability is required.

• Low-battery failsafe is required.

• Communication-loss recovery is required.

• Geofence-breach failsafe is required.

• Emergency recall capability is required.

• The operator must be able to safely abort the mission if required.

• The drone must be designed for safe operation inside confined indoor spaces.

19. Penalties

Violation	Penalty
Landing outside designated landing area	–10 per drone
Geofence / mission boundary breach	–20 per instance
Repeated geofence breach by same drone	Additional –20 per drone
Manual input or reset	–50 per instance
Crash	–50 per crash

Total penalties may normally be capped at 150 marks. Safety-critical violations can lead to mission termination or disqualification.

20. Design Review — 200 Marks

Design Review Area	Marks
Introduction, Team Composition & Domain Diversity	10
Overall Solution Architecture & Indoor Autonomy Strategy	25
Technical Approach, Drone Design & Indigenisation Strategy	20
GPS-Denied Navigation & Confined-Space Flight	30
Indoor Mapping Quality & Mission Planner Display	30
Survivor Detection, Localisation & Tagging Accuracy	35
Autonomous Mission Execution & Time Efficiency	20
Ground Control Station, Mission Map & Survivor Status Reporting	15
Fail-Safe Features & Safety	15
TOTAL	**200**

21. Business Strategy — 200 Marks

• Problem Understanding & Real-World Relevance — 30

• Target Users, Customers & Beneficiaries — 20

• Market Sizing & Deployment Potential — 30

• Business Model & Revenue Approach — 30

• Expenditure Breakdown & Resource Planning — 20

• Funds Raised, Sponsorships & Resource Mobilisation — 20

• Competitive Advantage & Differentiation — 20

• Go-to-Market Strategy & Partnership — 20

• Regulatory, Safety & Adoption Readiness — 10

22. Overall Competition Structure

Component	Maximum
Design Review Presentation	200
Business Strategy Presentation	200
Pre-Flight Inspection	Pass / Fail
Final Mission (Flight)	600
TOTAL	**1000**

23. Recommended Technical Priority

Based on the official scoring, the most important engineering priorities are:

1. Survivor detection + grid localisation — highest individual Final Mission category: 240 marks.

2. Accurate 2D grid mapping — 220 marks.

3. GPS-denied localisation + navigation — foundational because localisation quality affects both mapping and survivor tagging.

4. Autonomous obstacle avoidance — essential for safe confined-space operation.

5. Real-time integration — all mapping, survivor localisation and GCS updates must occur during flight.

> **Key takeaway:** AirMouse is not simply an autonomous-drone flying problem. It is an integrated problem of GPS-denied localisation, indoor exploration, mapping, survivor detection, grid-level survivor localisation, autonomous navigation and real-time mission reporting.

Source basis: NIDAR-26-27-Rulebook-Ver_2_1.pdf and Mission Brief - NIDAR AirMouse (1).pdf, provided in the conversation. Where the document does not specify an algorithm or an exact scoring normalisation formula, this PDF does not invent one.